

FunEat: An Interactive Tableware for Improving Eating Habits in Children

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ABSTRACT

Good eating habits are important for growing children. However, many children have anorexia and picky eating behaviors, causing a series of problems such as malnutrition and decreased resistance to illness. Our research focuses on guiding children to develop healthy eating habits through interactive animation and gamified design. We designed an intelligent dinner plate system, composed of a gravity-sensing dinner plate and a micro projection device, which can provide guidance through interactive animation when children eat. We conducted tests of the system with children aged 5-7 with picky eating behaviors. The results showed that the system had a positive effect on improving their eating habits.

CCS CONCEPTS

• Human-centered computing; • Human computer interaction (HCI);

KEYWORDS

children's habits, balanced diet, tableware, game design, interactive animation

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1 INTRODUCTION

Picky eating behaviors mean that children only choose one or a few foods they like when eating. This limited choice can lead to nutritional problems and is not conducive to healthy development. The reasons for these problems are twofold: children lack awareness about what constitutes balanced nutrition and parents lack effective methods to guide their children.

Some studies have attempted to address this problem through the design of tableware. Sangita Ganesh et al. used image recognition and projection systems to provide rewards when children finish their meal [2]. Azusa Kadomura et al. created an induction fork and mobile application to inspire children to eat correctly through the use of stories [6]. Inspired by the existing research, we integrate interactive animation into eating behavior to help children reduce picky eating.

In the present study, we developed an intelligent interactive system, composed of a pressure-sensing dinner plate and a micro projection device (Figure 1). It interacts with children through the form of simulation games, in which children are required to feed their pets by eating multiple types of food during the meal. The pets also judge the children's eating based on the weight of the plate and provide reminders. We developed a prototype of the plate and tested it with four children with picky eating behaviors to help us understand any problems with the system. In the present paper, we provide an overview of related research, the design of our systems and the results of the tests with children. We also summarize future research directions. We believe that integrating interactive animation and gamified design into the process of eating is an effective way to help children reduce picky eating behaviors and develop healthy eating habits.

2 RELATED WORK

In the field of human-computer interaction (HCI), there have been many research projects that integrate smart sensors into children's eating behaviors to encourage children to eat healthily. There are two main research directions.

Edutableware embedded sounds into a fork and cup to encourage healthy food choices [5]. Similarly, ChilDish used sensors placed around a cup and dish to make sounds corresponding to the drawings on the tableware, with the intention of encouraging children



Figure 1: Tableware system

to make healthy food choices through playful feedback [7]. Natasha Randall et al. designed a magic plate and robotic companion that reinforce children's healthy choices by lighting up an LED and playing music [10]. These previous studies mainly focused on the use of light and sound to interact with children during their meals and encourage healthy food choices.

Nora Ptakauskaite et al. used collective imagery weaves to study children's attitudes towards healthy eating and video games [1]. The aim of Sangita Ganesh et al.'s research was to encourage pre and primary school children to make healthier choices through digitally enhancing the food they dislike [2]. Azusa Kadomura et al. designed the Sensing Fork and Hungry Panda game for a young child who is a picky eater and/or easily distracted during mealtime [6]. Ying-Yu Chen et al. presented an evaluation of three prototype tangible user interfaces (TUIs) for preschoolers during mealtime to research the effectiveness of these novel tangibles in changing children's behaviors [8]. Yeong Rae Joi et al. presented a multi-connected tableware system, including a smart tray and a spoon, for changing children's behavior to eat more vegetables willingly [9]. These previous studies focused on using tangible feedback and gamification to cultivate healthy eating habits.

From these advances in the study of children's healthy eating habits, we found that real-time feedback and gamification have a positive effect on promoting children's eating habits. We finally chose interactive animation. One of the reasons is the previous research has not adopted such a format, which is relatively novel. Another reason is interactive animation has the characteristic of strong real-time feedback. We hope that we can establish an animated character with vitality and connect it with children's behavior. Based on previous research, we further study the combination of interactive animation and projection to support children's healthy eating habits.

3 PROTOTYPE DESIGN

FunEat consists of a smart dinner plate, micro projector and laptop. The bottom sections of the four grids of the plate are respectively equipped with a pressure sensor (RP-C18.3-LT) based on the Arduino platform (Figure 2). This can detect the change of food weight during the meal and immediately transmit the data to the computer. The change of the value in the serial port of Arduino can be seen. Serial communication between Arduino and the processing unit is established so that data can be transmitted between hardware

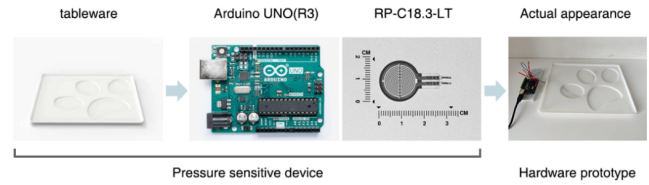


Figure 2: Hardware prototype

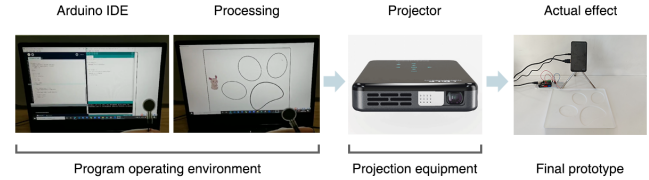


Figure 3: Final prototype

and software. The processing unit obtains the change of the food weight value from Arduino, generates an interactive animation according to the designed conditions and uses the audio player to play a pre-prepared audio on the computer. The micro projector is placed in front of the plate and will project an animation onto the plate (Figure 3).

FunEat is designed to cultivate children's healthy eating habits. To achieve this goal, we clarified two concepts as the key points of this design. One is the simulation game: children need to feed their pets by eating different kinds of food, which can motivate them to eat healthily. The second is instant feedback: pets will exhibit various behaviors according to the children's eating behavior, which can make children focus on their eating. We used the form of projection to project the animated picture on the plate, so that the picture can directly interact with the food. This is intended to be more interesting to the children and have a stronger guiding effect.

4 INTERACTION AND GAMIFICATION

We designed a game that can be projected onto the dinner plate and interact with children in real time during the meal. In the game, children need to feed a pet, whose growth rate depends on the child's completion and level of picky eating during each meal. A lower picky eating level leads to higher completion and faster growth of the pet.

During the meal, the pressure sensor detects the difference in pressure in different areas of the dinner plate. If the difference is within a certain range, it indicates that the child's eating is normal. The pet will actively move to the food area where the child is eating, which will encourage the child to continue to eat. If the difference exceeds a certain value, it indicates that the child frequently eats only certain foods and is judged to have picky eating behavior. The pet will move to the food area that the child does not like and, through actions and sounds, will remind the child to eat a balanced diet. Children can adjust their eating habits by observing the growth of their pet and the prompts in the game.

The entire game design is based on the development model. Children are required to complete the meal goals according to the

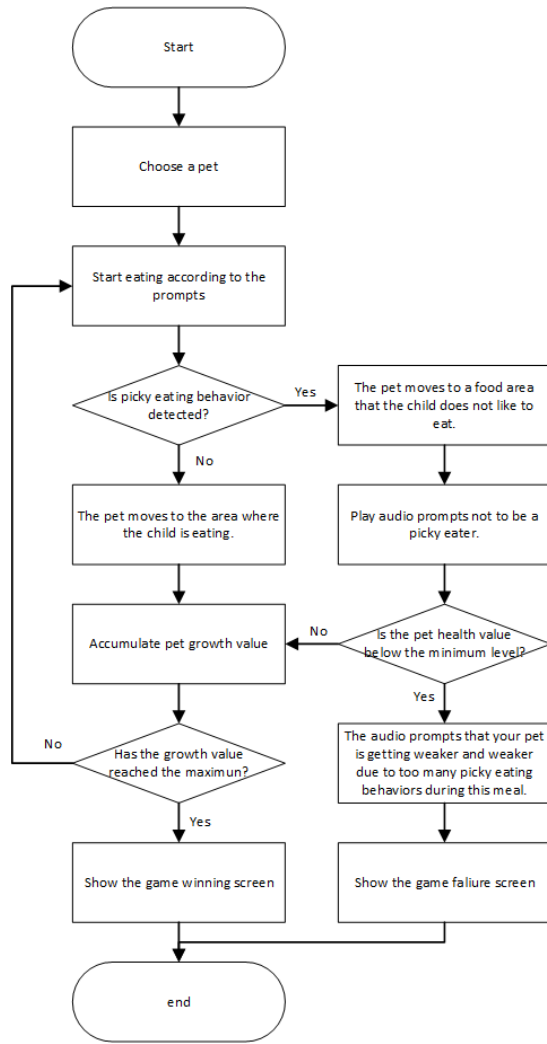


Figure 4: Game flow chart

prompts to help the animals grow. Before the meal starts, the child can choose the animals to be fed on the farm. After this, the game makes a sound to prompt the child to start eating. During the meal, the upper left corner of the tableware simultaneously displays the current growth value and health value of the pet, which will continue to change during the meal. The health value represents children's picky eating behavior: the less that picky eating behaviors occur, the higher the health value. The growth value can be accumulated and when it reaches a maximum, the animal will be upgraded. Children can know whether they have picky eating behaviors according to the changes in the health value and growth value and can make corresponding adjustments according to the animation prompts, so that the animal can grow healthily and quickly. After the meal is over, the word "Win" appears on the tableware, to indicate that the child has displayed good eating behavior and is not picky.

The game flow chart is shown in Figure 4, operation flow chart is shown in the Figure 5 and the interactive animation in Figure 6

5 PRIMARY USER TEST

We tested the system with four children with picky eating behaviors (2 females, 2 males; age range 5 to 6 years old; mean age 5.5 years) who do not like vegetables and fruits. We recruited the children by contacting a school. We informed the children and their parents about the study and obtained their consent. The aim of the study was to test the effect of interactive tableware on helping children develop a balanced diet.

Before the test, we prepared different types of foods on the plate, including meat, vegetables and fruits. The children were told that they could interact with the animal on their plate, pay attention to its cues and feed it by eating. Parents accompanied the whole testing process in order to simulate a daily meal scene. We recorded the test with a fixed camera to reduce the impact of the presence of researchers on the children (Figure 7).

We explored children's and parents' perceptions through a brief questionnaire composed of four and five questions on a five-point Likert scale. We also asked the children and their parents to describe how they felt during the test.

6 RESULTS AND DISCUSSION

Our results clearly showed that children can be guided to improve their eating habits by interacting with projected animation. On the questionnaire for children (Table 1), the average score of children was 4.25 for 'how interesting the food plate is', 4.5 for 'whether they are willing to help small animals grow up' and 4.25 for 'how they feel about the result'. These findings showed that projected animations and simulation games are attractive to children, who are willing to take actions according to the plate's guidance to achieve their goals. Parents scored 4.5 for both 'overall impression of the plate' and 'improvement of children's eating habits after the plate compared to normal' (Table 2). These findings indicated that parents had a positive view of the plate and believed that it could have a positive effect on children's eating habits. The question 'does the use of the system affect communication among family members?' scored 4.5, as parents felt they could maintain communication with their children even with animations and prompts.

While most of the feedback was positive, children and parents scored low on the question of whether animation has an impact on eating, at 4 and 3.25 respectively. One of the parents thought that this form of tableware will make children pay more attention to the projected pictures, leading to distraction, then slower eating speed.

In the one-on-one interviews, one child said he was happy to see the rabbit grow up and the word 'Win' on his plate. One mother told us it was the first time her child, who rarely eats vegetables, volunteered to eat every vegetable on his plate. These findings suggest that FunEat can give children a better eating experience. It innovatively integrates interactive animation into the eating process and helps children develop habits via guidance rather than commands, making them feel that eating is very interesting.

7 CONCLUSION AND FUTURE WORK

In the present paper we have introduced FunEat, a smart dinner plate system composed of a pressure-sensing dinner plate and a micro projection device. It can help children develop healthy eating habits through interactive animations while eating. The interactive

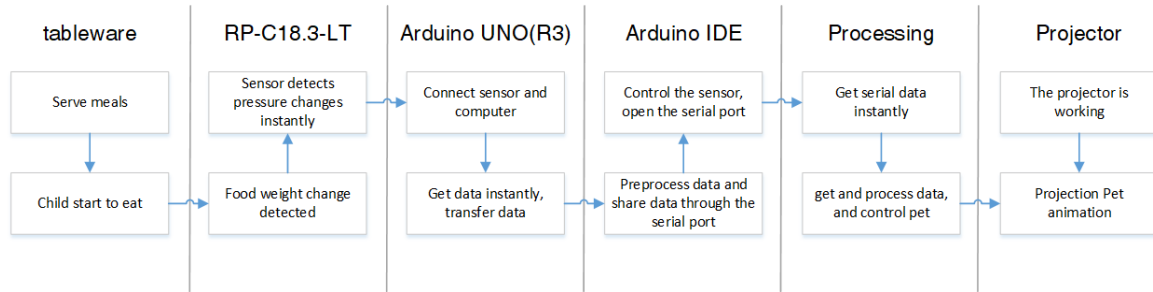


Figure 5: Operation flow chart

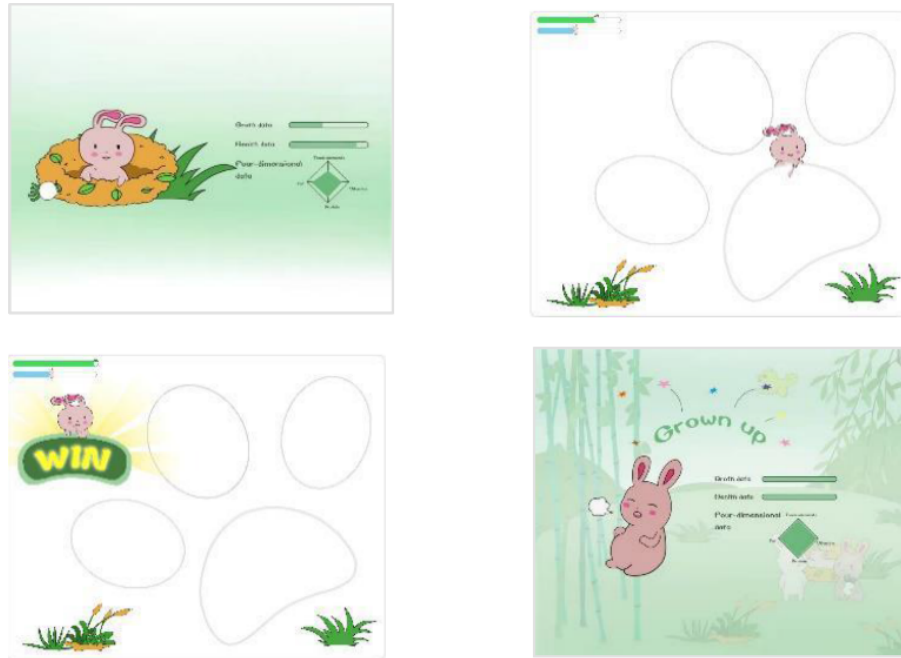


Figure 6: Interactive animation

Table 1: Results of the questionnaire for children

Questions	BoyA	BoyB	GirlC	GirlD	Avg
I. Do you find it interesting to eat on this plate? (1: boring – 5: interesting)	4	4	5	4	4.25
II. Do you think animation will interfere with your eating? (1: much – 5: not at all)	4	3	4	5	4
III. Are you willing to help animals grow up with good eating habits? (1: unwilling – 5: willing)	4	4	5	5	4.5
IV. How do you feel when you see the rabbit grow up? (1: negative – 5: positive)	4	4	5	4	4.25

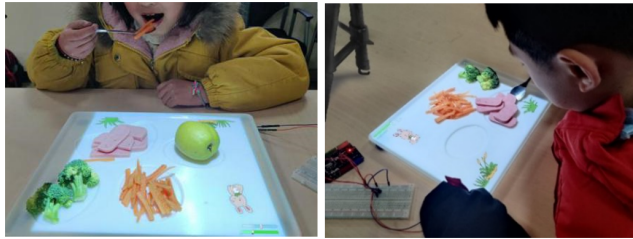
system combines a simulation game with instant feedback when eating.

Our study showed that FunEat has the potential to develop healthy eating habits in children. However, there are some limitations. One is that the current gameplay is relatively simple, which requires richer animations and gameplay to keep children interested. The other is that a small number of parents believed

that the projected pictures may distract their children's attention when eating. To address this problem, we will work to optimize the picture design to reduce the impact of the picture on the food. We will continue to improve FunEat and conduct controlled trials to explore its effectiveness in improving children's eating habits.

Table 2: Results of the questionnaire for parents

Questions	ParentA	ParentB	ParentC	ParentD	Avg
I. What is your overall impression of our system? (1: negative – 5: positive)	4	4	5	5	4.5
II. Do you think animation will interfere with your child's eating? (1: much – 5: not at all)	3	2	4	4	3.25
III. Is it easy for children to understand the rules of the game? (1: hard – 5: easy)	5	5	3	4	4.25
IV. Do you think the system will affect the communication between family members? (1: much – 5: not at all)	4	5	4	5	4.5
V. Do you think your child's eating habits have improved after using the plate? (1: not at all – 5: much)	4	4	5	5	4.5

**Figure 7: Children's test**

REFERENCES

- [1] Nora Ptakauskaite, Priscilla Chueng-Nainby, and Helen Pain. 2016. Supporting Social Innovation in Children: Developing a Game to Promote Health Eating. In Proceedings of the The 15th International Conference on Interaction Design and Children (IDC '16). Association for Computing Machinery, New York, NY, USA, 688–693.
- [2] Sangita Ganesh, Paul Marshall, Yvonne Rogers, and Kenton O'Hara. 2014. Food-Works: tackling fussy eating by digitally augmenting children's meals. In Proceedings of the 8th Nordic Conference on Human-Computer Interaction: Fun, Fast, Foundational (NordiCHI '14). Association for Computing Machinery, New York, NY, USA, 147–156.
- [3] Eunkyung Jo, Hyeonseok Bang, Myeonghan Ryu, Eun Jee Sung, Sungmook Leem, and Hwajung Hong. 2020. MAMAS: Supporting Parent-Child Mealtime Interactions Using Automated Tracking and Speech Recognition. Proc. ACM Hum-Comput. Interact. 4, CSCW1, Article 066 (May 2020), 32 pages.
- [4] Ying-Yu Chen, Ziyue Li, Daniela Rosner, and Alexis Hiniker. 2019. Understanding Parents' Perspectives on Mealtime Technology. Proc. ACM Interact. Mob. Wearable Ubiquitous Technol. 3, 1, Article 5 (March 2019), 19 pages.
- [5] Azusa Kadomura, Koji Tsukada, and Iitiro Siiro. 2013. EducaTableware: computer-augmented tableware to enhance the eating experiences. In CHI '13 Extended Abstracts on Human Factors in Computing Systems (CHI EA '13). Association for Computing Machinery, New York, NY, USA, 3071–3074.
- [6] Azusa Kadomura, Cheng-Yuan Li, Koji Tsukada, Hao-Hua Chu, and Iitiro Siiro. 2014. Persuasive technology to improve eating behavior using a sensor-embedded fork. In Proceedings of the 2014 ACM International Joint Conference on Pervasive and Ubiquitous Computing (UbiComp '14). Association for Computing Machinery, New York, NY, USA, 319–329.
- [7] Seo Yoon Han and Eun Ji Kang. 2017. ChildDish: The Smart Plate and Cup for Children. In Proceedings of the Companion of the 2017 ACM/IEEE International Conference on Human-Robot Interaction (HRI '17). Association for Computing Machinery, New York, NY, USA, 391–392.
- [8] Ying-Yu Chen, Jason Yip, Daniela Rosner, and Alexis Hiniker. 2019. Lights, Music, Stamps! Evaluating Mealtime Tangibles for Preschoolers. In Proceedings of the Thirteenth International Conference on Tangible, Embedded, and Embodied Interaction (TEI '19). Association for Computing Machinery, New York, NY, USA, 127–134.
- [9] Joi, Y. R., Jeong, B. T., Kim, J. H., Park, J., Cho, J., Seong, E., Bae, B.-C. and Cho, J. D. Interactive and Connected Tableware for Promoting Children's Vegetable-Eating and Family Interaction. In Proceedings of the Proceedings of the The 15th International Conference on Interaction Design and Children - IDC '16 (2016).
- [10] Randall, N., Joshi, S. and Liu, X. Health-e-Eater. In Proceedings of the Companion of the 2018 ACM/IEEE International Conference on Human-Robot Interaction (2018).
- [11] Cheng Zheng, Caowei Zhang, Xuan Li, Xin Liu, Chuqi Tang, Guanyun Wang, Cheng Yao, Fan Zhang, Wenjie Xu, and Fangtian Ying. 2017. Toon-chat: a cartoon-masked chat system for children with autism. In ACM SIGGRAPH 2017 Posters (SIGGRAPH '17). Association for Computing Machinery, New York, NY, USA, Article 50, 1–2.